Advanced Labs I （AY2023 ）

Theme			
X-Ray diffraction			
Dates of Experiments	Oct. 9, Oct 16, Oct 23	Report Submission Date	Oct 28
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Advanced Labs - Theme A: X-ray Diffraction

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October 28, 2023

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1 Experiment Objectives

The objective of our experiments was to identify simple crystals using two main methods: the Laue method and the Diffractometer method.

1.1 Week 1

In the first week we focused on the Laue Method to identify the indices of Laue spots generated by the crystals, namely LiF(001), Si(001) and Si(111), we then analyzed the symmetry present with these Laue patterns.

The Laue Photo was generated by firing an X-ray beam directly perpendicular to a thin flat piece of the crystal. We identified the symmetry of the Laue patterns if any was present by comparing the Laue spots on the photo to the poles present on the standard projection of a cubic crystal of miller index (001).

We only calculated the miller indices for the LiF sample because Si had a different crystalline structure to LiF and we only had a (001) standard projection.

1.2 Week 2

In the second week, we used the diffractor method to identify various crystals by changing the angle at which the X-ray hits the crystal tested and measuring the corresponding intensity of the reflected X-ray. Using the results and software we get a Diffraction Pattern which is a graph of different intensity peaks. To identify each crystal type, the angle of each intensity peak is used. Using the results we derive the indexing patterns using the formula:

$$2d_{hkl} \sin\theta = \lambda$$

and the derivation of the lattice constant a using:

$$\frac{\sin\theta^2}{h^2 + k^2 + l^2} = \frac{\lambda^2}{4a^2} = const$$

We start the experiment with a known sample of Si-crystal powder to ensure that everything is working properly and then move on to two unknown samples of Cu and Mo.

2 Instruments and Methods

2.1 Week 1

For week one, we used an X-ray beam, using the X-ray machine shown in figure 1, that was directed onto a perpendicular thin crystal of LiF (001). Inside the machine, the X-ray is produced by a cathode-filament tube, shown in Fig 2, which produces a white, continuous X-ray with a variable wavelength λ. The crystal diffracted the X-ray onto a CMOS camera 3cm away which we used to take a Laue photo. We analyzed the photo to identify Miller indices and symmetry.



Figure 1: X-Ray machine used for week 1

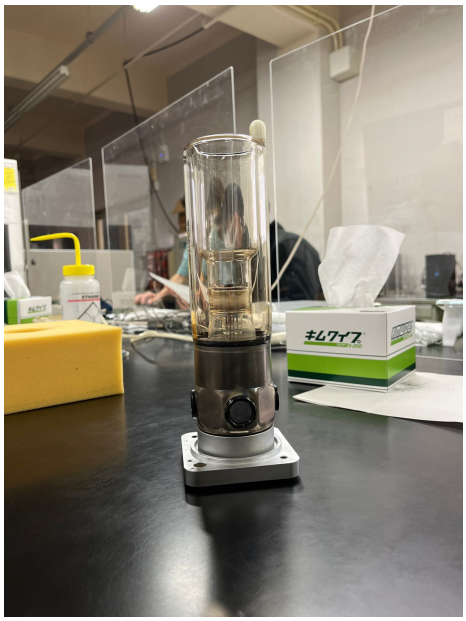


Figure 2: Cathode Tube

We set the voltage value to 30kV, current to 20mA, exposure time to 5 seconds and the number of shots to 3 using the control software.

We repeated this for the samples of Si (001) and Si (111), however, we had to change the voltage to 40kV for the Si(111) because the resulting picture was not clear.

2.2 Week 2

In week 2, we measured the diffraction pattern of a sintered si plate for θ range between 20° and 150°. We used the machine shown in fig 3



Figure 3: Week 2 X-ray Machine

This machine produces a monochromatic, characteristic X-ray with a constant wavelength. The x-ray passes through soller slits to make sure that the x-rays are parallel to each other. It also passes through a Ni filter to monochromize the radiation. We set the power to 40kV and the current to 10mA.

The resulting diffraction intensity is detected by a counter. We used specialized software to identify peaks and form them into a list Then we analyzed the results using Excel and found the Miller indices and structure factor which will be discussed in detail later. We repeated the experiment for two unknown crystals.

3 Results and Analysis

3.1 Week 1

From the X-ray machine, we got Laue pictures for LiF(001) shown in Fig 4, Si(001) in Fig 5, and Si(111) in Fig 6.

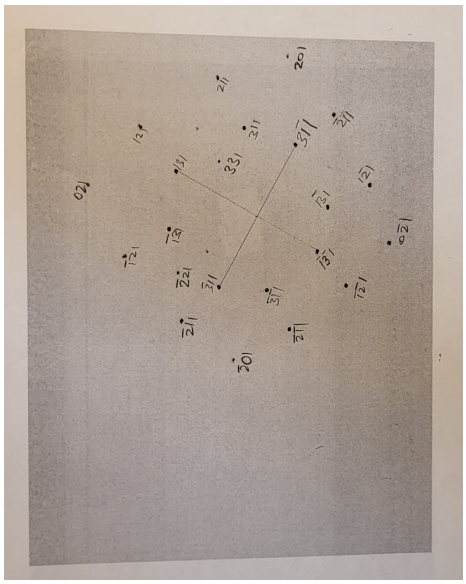


Figure 4: LiF(001) Laue Photo with symmetry lines

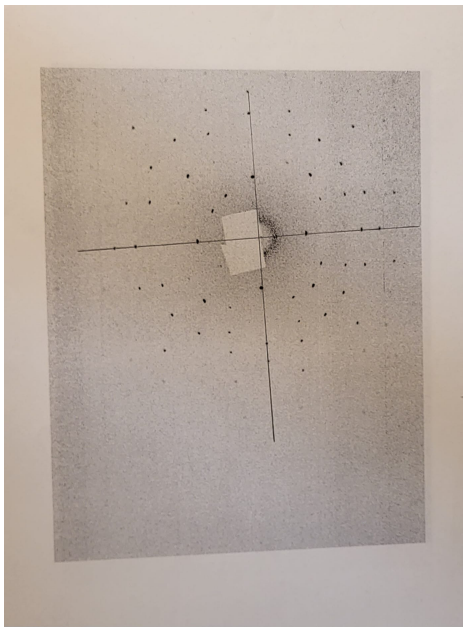


Figure 5: Si(001) Laue Photo with symmetry lines

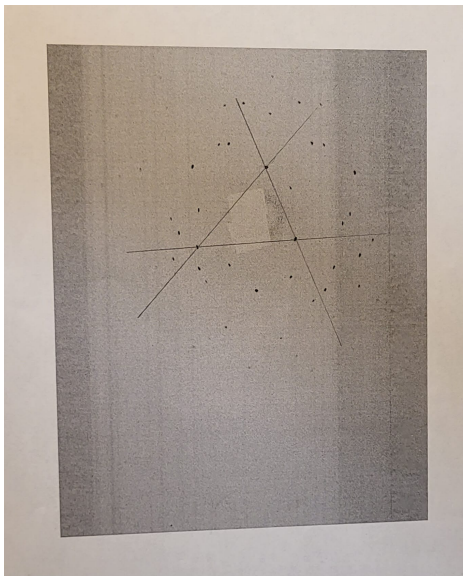


Figure 6: Si(111) Laue Photo with symmetry lines

We can note the 4-fold symmetry for LiF(001) in fig 4, 4-fold symmetry for Si(001) in fig 5, and Tri-fold symmetry for Si(111) in fig 6. A four-fold symmetry means that if we are to rotate the photo by 90° we would obtain the same image, and 120° for a tri-fold symmetry.

We used a (001) standard projection, shown in Fig 7, to index the Laue spots. We overlayed the laue photograph on top of the standard projection and then used a backlight to highlight the spots. We then used a Stereographic ruler, shown in Fig 8, to match each of the laue spots with a pole on the standard projection, which allows us to index the Laue spot. We only index the Laue spots for LiF(001) because, as mentioned before, we only have a (001) projection and the crystal structure of Si is too complicated.

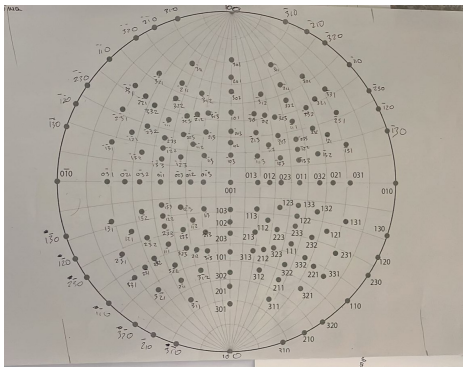


Figure 7: (001) Standard Projection



Figure 8: Stereographic ruler

3.2 Week 2

From the experiment of the second week, we got a diffraction pattern and Analyzed the patterns using the software and got the diffraction peak graph for sample 1 shown in Fig 9, the diffraction of sample 2 shown in Fig 10 and the diffraction pattern of sample 3 shown in Fig 11.

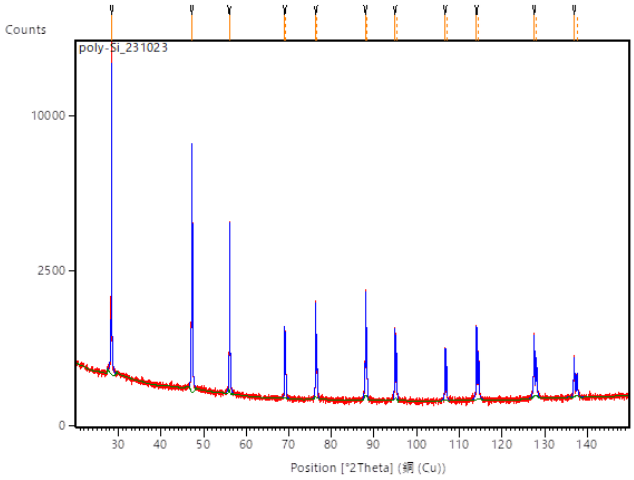


Figure 9: Diffraction peaks of Si sample

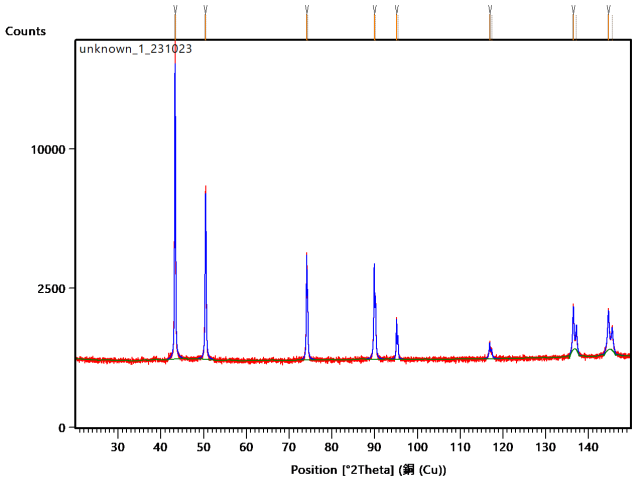


Figure 10: diffraction peaks of unknown sample 1

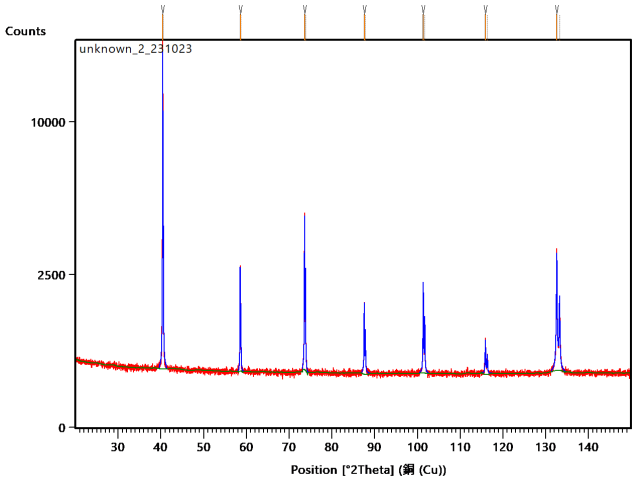


Figure 11: diffraction peaks of unknown sample 2

Using the data analyzed by the software we fill the table of data for the Si sample, unknown sample 1 and unknown sample shown in table 1, table 2 and table 3 respectively.

λ(Å)	Number	$2\theta_i$	θ_i	radian	$\sin^2\theta_i$	$\frac{\sin^2\theta_i}{\sin^2\theta_1}$	仮定したS_i	$\frac{\sin^2\theta_i}{\sin^2\theta_1}S_1$	整数化したS_i	$\frac{\lambda^2}{4a^2}$	a(Å)	$\frac{\cos^2\theta_i}{\sin\theta_i}$	hkl	d_{hkl}(Å)	実測値	文献値	誤差(%)
1.54051		28.3991	14.2	0.24783	0.06017	1	3	3	3	0.020057	5.43874	3.83134	1,1,1	3.14023	5.4306	5.428	0.000479
1.54433		28.4712	14.236	0.24846	0.06047	1.00498		3.014937	3	0.020157	5.4387	3.82062	1,1,1	3.13245			
1.54051		47.2673	23.634	0.41249	0.16071	2.67085		8.012559	8	0.020089	5.43447	2.09358	0,2,2	1.92149			
1.54433		47.3919	23.696	0.41357	0.16151	2.68414		8.052419	8	0.020189	5.43445	2.0864	0,2,2	1.91672			
1.54051		56.0862	28.043	0.48944	0.22103	3.67326		11.01979	11	0.020093	5.43385	1.65691	3,1,1	1.63846			
1.54433		56.2379	28.119	0.49077	0.22213	3.69154		11.07461	11	0.020193	5.43383	1.65047	3,1,1	1.6344			
1.54051		69.0945	34.547	0.60296	0.32159	5.34445		16.03335	16	0.020099	5.43308	1.19632	4,0,0	1.35835			
1.54433		69.2907	34.645	0.60468	0.32319	5.37105		16.11315	16	0.020199	5.43305	1.19054	4,0,0	1.35498			
1.54051		76.3422	38.171	0.66621	0.38194	6.34745		19.04236	19	0.020102	5.43268	1.00008	3,3,1	1.24641			
1.54433		76.5662	38.283	0.66817	0.38384	6.37803		19.1371	19	0.020202	5.43266	0.99453	3,3,1	1.24332			
1.54051		87.9974	43.999	0.76792	0.48253	8.01914		24.05742	24	0.020105	5.43224	0.74495	4,2,2	1.10892			
1.54433		88.2726	44.136	0.77032	0.48493	8.05903		24.17709	24	0.020205	5.43222	0.73966	4,2,2	1.10617			
1.54051		94.9204	47.46	0.82834	0.54289	9.02224		27.06671	27	0.020107	5.43203	0.6204	3,3,3	1.04545			
1.54433		95.2312	47.616	0.83105	0.54559	9.06713		27.2014	27	0.020207	5.432	0.6152	3,3,3	1.04286			
1.54051		106.6814	53.341	0.93097	0.64352	10.6948		32.08428	32	0.02011	5.43159	0.44437	4,4,0	0.96023			
1.54433		107.0648	53.532	0.93432	0.64673	10.748		32.24391	32	0.02021	5.43156	0.43929	4,4,0	0.95785			
1.54051		114.0659	57.033	0.99541	0.70389	11.698		35.09409	35	0.020111	5.43144	0.35293	5,3,1	0.91813			
1.54433		114.5062	57.253	0.99926	0.7074	11.7562		35.2687	35	0.020211	5.43141	0.3479	5,3,1	0.91586			
1.54051		127.521	63.761	1.11283	0.80453	13.3704		40.11133	40	0.020113	5.43118	0.21793	6,2,2	0.85879			
1.54433		128.1016	64.051	1.1179	0.80853	13.437		40.3109	40	0.020213	5.43116	0.21294	6,2,2	0.85667			
1.54051		136.8805	68.44	1.19451	0.86496	14.3749		43.12463	43	0.020115	5.43087	0.14519	5,3,3	0.82825			
1.54433		137.607	68.804	1.20085	0.86927	14.4464		43.33922	43	0.020216	5.43084	0.14022	5,3,3	0.82619			

Table 1: Table of data derived from Diffraction Pattern of Si crystal

λ(Å)	Number	$2\theta_i$	θ_i	radian	$\sin^2\theta_i$	$\frac{\sin^2\theta_i}{\sin^2\theta_1}$	仮定したS_i	$\frac{\sin^2\theta_i}{\sin^2\theta_1}S_1$	整数化したS_i	$\frac{\lambda^2}{4a^2}$	a(Å)	$\frac{\cos^2\theta_i}{\sin\theta_i}$	hkl	d_{hkl}(Å)	実測値	文献値	誤差(%)
1.54051		43.3009	21.65	0.37787	0.13612	1	3	3	3	0.045273	3.61606	2.3415	1,1,1	2.08784	3.615	3.6153	8.3E-05
1.54433		43.414	21.707	0.37886	0.1368	1.00498		3.014934	3	0.045399	3.61604	2.33386	1,1,1	2.08267			
1.54051		50.434	25.217	0.44012	0.18152	1.33351		4.000544	4	0.045379	3.61582	1.92111	2,0,0	1.80801			
1.54433		50.5681	25.284	0.44129	0.18242	1.34015		4.020445	4	0.045605	3.6158	1.91423	2,0,0	1.80353			
1.54051		74.115	37.058	0.64678	0.36315	2.66766		8.003576	8	0.045393	3.61525	1.06681	2,2,0	1.27826			
1.54433		74.3301	37.165	0.64865	0.36495	2.68113		8.043986	8	0.045619	3.61524	1.06121	2,2,0	1.27509			
1.54051		89.9192	44.96	0.78469	0.48929	3.69808		11.00423	11	0.04539	3.61537	0.7096	3,1,1	1.09012			
1.54433		90.2039	45.102	0.78718	0.50178	3.68633		11.05899	11	0.045616	3.61535	0.70834	3,1,1	1.08741			
1.54051		95.1311	47.566	0.83018	0.54472	4.00177		12.00532	12	0.045393	3.61526	0.61687	2,2,2	1.0437			
1.54433		95.443	47.722	0.8329	0.54743	4.02168		12.06505	12	0.045619	3.61524	0.61168	2,2,2	1.04111			
1.54051		116.914	58.457	1.02027	0.72633	5.33596		16.00788	16	0.045395	3.61517	0.32112	4,0,0	0.90384			
1.54433		117.379	58.69	1.02433	0.72994	5.36251		16.08753	16	0.045621	3.61515	0.3161	4,0,0	0.9016			
1.54051		136.479	68.239	1.191	0.86256	6.3368		19.01039	19	0.045398	3.61507	0.14799	3,3,1	0.8294			
1.54433		137.198	68.599	1.19727	0.86685	6.36833		19.10498	19	0.045624	3.61505	0.14301	3,3,1	0.82735			
1.54051		144.68	72.34	1.26257	0.90797	6.6704		20.01119	20	0.045398	3.61505	0.09658	4,2,2	0.8084			
1.54433		145.586	72.793	1.27047	0.91249	6.70359		20.11076	20	0.045624	3.61503	0.09161	4,2,2	0.80639			

Table 2: Table of data derived from Diffraction Pattern of unknown crystal 1

λ(Å)	Number	$2\theta_n$	θ_n	radian	$\sin^2\theta_n$	$\frac{\sin^2\theta_n}{\sin^2\theta_1}$	仮定したS	$\frac{\sin^2\theta_n}{\sin^2\theta_1} S_1$	整数化したS	$\frac{\lambda^2}{4a^2}$	a(Å)	$\frac{\cos^2\theta_m}{\sin\theta_m}$	hkl	d_{hkl}(Å)	実測値	文献値	誤差(%)
1.54051		40.4348	20.217	0.35286	0.11943	1	2	2	2	0.059714	3.15208	2.54808	1.1.0	2.08784	3.1471	3.1466	0.000159
1.54433		40.5397	20.27	0.35378	0.12002	1.00498		2.009954	2	0.060011	3.15206	2.54004	1.1.0	2.08267			
1.54051		58.5551	29.278	0.51099	0.23916	2.00256		4.005113	4	0.05979	3.15006	1.55578	2.0.0	1.80801			
1.54433		58.7148	29.357	0.51238	0.24035	2.01252		4.025042	4	0.060088	3.15005	1.5495	2.0.0	1.80353			
1.54051		73.6133	36.807	0.6424	0.35894	3.0055		6.011009	6	0.059823	3.14919	1.07001	2.1.1	1.27826			
1.54433		73.8265	36.913	0.64426	0.36073	3.02046		6.040917	6	0.060121	3.14917	1.06438	2.1.1	1.27509			
1.54051		87.5653	43.783	0.76415	0.47876	4.00879		8.017561	8	0.059845	3.14862	0.75332	2.2.0	1.09012			
1.54433		87.5793	43.79	0.76427	0.47888	4.0098		8.019606	8	0.05986	3.15603	0.75305	2.2.0	1.08741			
1.54051		87.8385	43.919	0.76654	0.48114	4.02873		8.057455	8	0.060143	3.14862	0.74802	2.2.0	1.0437			
1.54433		87.8526	43.926	0.76666	0.48126	4.02976		8.059514	8	0.060158	3.14821	0.74775	2.2.0	1.04111			
1.54051		101.376	50.688	0.88467	0.59862	5.01243		10.02486	10	0.059862	3.14817	0.51877	3.1.0	0.90384			
1.54433		101.724	50.862	0.88771	0.6016	5.03737		10.07475	10	0.06016	3.14815	0.51364	3.1.0	0.9016			
1.54051		115.911	57.956	1.01152	0.71849	6.01611		12.03222	12	0.059874	3.14785	0.33211	2.2.2	0.8294			
1.54433		116.368	58.184	1.0155	0.72207	6.04605		12.0921	12	0.060172	3.14784	0.32708	2.2.2	0.82735			
1.54051		132.596	66.298	1.15712	0.83841	7.02023		14.04046	14	0.059886	3.14753	0.17648	3.2.1	0.8084			
1.54433		133.249	66.624	1.16281	0.84258	7.05516		14.11032	14	0.060184	3.14752	0.17149	3.2.1	0.80639			

Table 3: Table of data derived from Diffraction Pattern of unknown crystal 2

The 2θ represents twice the angle of incidence at which X-ray peaks occur. From this, we derive the rest of the data.

We derive the lattice parameter, "a", by the formula:

$$a = \sqrt{s * d_{hkl}^2}$$

We plotteed the lattice parameter against the extrapolation function

$$\frac{\cos\theta^2}{sin\theta}$$

We extrapolated the value of 2θ up to 180° the results are shown in Figs 12, 13, and 14 for sample Si, Unknown 1 and Unknown 2 respectively. the results show the relationship between a and exponential function values.

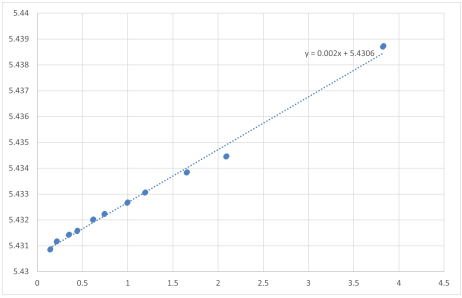


Figure 12: Si Sample graph

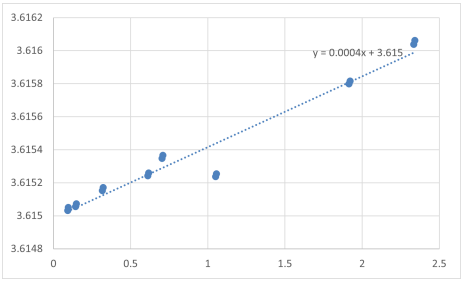


Figure 13: Unknown 1 Sample graph

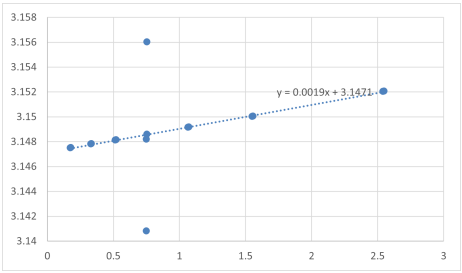


Figure 14: Unknown 2 Sample graph

We determine the most accurate lattice constant value by finding the y-intercept of the line of best fit of the points. We get xxx for Si, xxx for unknown 1 and xxx for unknown 2.

3.3 Structure Factor Calculation

Based on existing data we can identify the structure type of each sample as shown in table 4.

Sample	Structure Type
LiF	NaCl-Type
Si	Diamond-Type
Cu	Face-centered cubic
Mo	Body-centered cubic

Table 4: Structure type of samples table

The structure factor of LiF, F_{hkl}, is given by:

$$F_{hkl} = 4(f_{Na} + f_{Cl}) \text{ if } (h,k,l) \text{ are all even}$$
$$F_{hkl} = 4(f_{Na} - f_{Cl}) \text{ if } (h,k,l) \text{ are all odd}$$
$$F_{hkl} = 0 \text{ if } (h,k,l) \text{ are mixed}$$

So if the value of the indices are mixed then abscence will occur. For example, the following indices will vanish:

- 001
- 110
- 210
- 211

The structure factor of Cu is given by:

$$F_{hkl} = 4f \text{ if } (h,k,l) \text{ are unmixed}$$
$$F_{hkl} = 0 \text{ if } (h,k,l) \text{ are mixed}$$

Given that Cu has a face-centered lattice, we can assume that the following indices will experience systematic absence:

- 001
- 110
- 210
- 211

The structure factor of Mo is given by:

$$F_{hkl} = 0 \text{ if } h+k+l \text{ is odd}$$
$$F_{hkl} = 2f \text{ if } h+k+l \text{ is even}$$

Given that Mo has a body-centered lattice, we can assume that the following indices will experience systematic absence:

- 201
- 111
- 221
- 210

The structure factor of Si is given by:

$$F_{hkl} = 8f \text{ if } h + k + l = 4m$$

$$F_{hkl} = 4f(1 \mp \text{i}) \text{ if } h + k + l = 4m \pm 1$$

$$F_{hkl} = 0 \text{ if } h + k + l = 4m \pm 2$$

$$F_{hkl} = 0 \text{ if } (h, k, l) \text{ are mixed}$$

Given that Si has a diamond-type lattice, we can assume that the following indices will experience systematic absence:

- 001
- 200
- 211
- 210

4 Discussion

4.1 Week 1

To find out which indices lead to systemic absences, we can look at the Laue picture. If spots are missing on the Laue picture when compared to a normal projection, those are the spots where systemic absences happen. By comparing the Laue picture with a normal projection, we can see which spots are not there. We then use tools like a stereographic ruler and the standard projection to label those spots and find out which Miller indices are causing systemic absences.

Earlier, we noticed that the Laue pictures of Si (<001>) and LiF showed a pattern that repeated four times, while the Laue picture of Si(<111>) showed a pattern that repeated three times. This fits with what we know about how the structure of crystals can create different symmetry patterns. We can also check this against the research given to us about the tests we did in the last two weeks.

4.2 Week 2

From the data, we can confirm that the unknown samples 1 and 2 are of Cu and Mo respectively. The theoretical and experimental values are shown in Table 5.

Sample	Literature Value/Å	Measured Value/Å	Percentage Difference/%
Si	5.4282	5.4306	0.000479
Cu	3.6153	3.6153	0.000083
Mo	3.1466	3.1471	0.0000159

Table 5: Comparison of Literature and Measured Values

We can infer from the table that the differences between the theoretical and measured value are very minute and can be due to inconsistent intensity of x-ray due to voltage fluctuation or impurities within the crystal.

The summary of the process to identify the substances of the "unknown samples" is as follows:

1. Obtain the diffraction pattern of the sample in the 20° 150° 2 range using the X-ray specifications provided.
2. Match diffraction peaks to their (hkl) miller indices.
3. Find the angle of each peak using the dhkl spacing.
4. Compute sin2 for each peak.
5. Extract the lattice constant a from sin2 .
6. R2fine the lattice constant using the Week 1 Analysis extrapolation method.
7. Match the lattice constant with literature to identify the substance.